Data Valorization

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Business goal

Business problem:

- Business costs
- Revenue loss
- Safety risks

Business value:

- Cost reduction
- Increased efficiency and productivity
- Risk mitigation

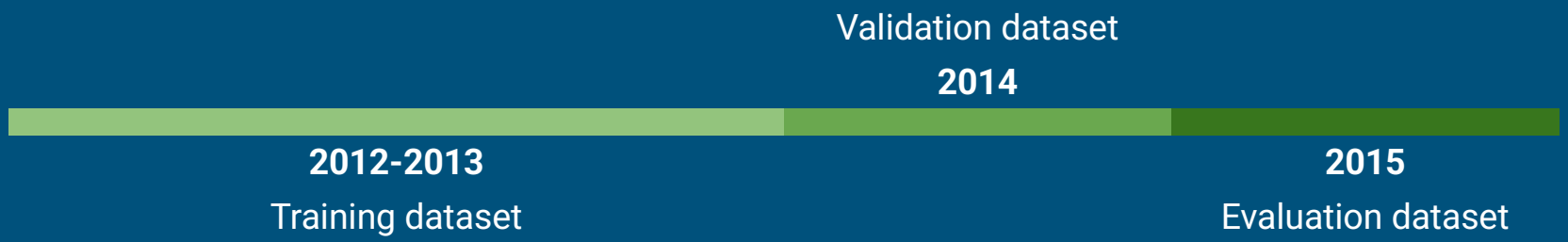
Target users:

- Energy providers
- Agriculture and farming operations
- Logistics, travelling, tourism, etc.

Technical goal

Provide an accurate estimation of the weather

- Train, validate, evaluate method
- Search for hyperparameters
- Data ordered by date



Data description

- Dataset name: **Seattle Weather**
- Number of records: **1641 1** (daily observations)
- Variables:

Quantitative variables:

- Date
- Precipitation (mm)
- Maximum temperature (°C)
- Minimum temperature (°C)
- Wind speed(km/h)

Qualitative variable:

- Weather type
  (rain, sun, snow, fog, drizzle)

Our dataset:

https://www.kaggle.com/datasets/ananthr1/weather-prediction

	date <chr>	precipitation <dbl>	temp_max <dbl>	temp_min <dbl>	wind <dbl>	weather <chr>
1	2012-01-01	0.0	12.8	5.0	4.7	drizzle
2	2012-01-02	10.9	10.6	2.8	4.5	rain
3	2012-01-03	0.8	11.7	7.2	2.3	rain
4	2012-01-04	20.3	12.2	5.6	4.7	rain
5	2012-01-05	1.3	8.9	2.8	6.1	rain
6	2012-01-06	2.5	4.4	2.2	2.2	rain

Data quality with univariate analysis

Statistical summary:

- The dataset contains 1,461 daily observations
- Precipitation is highly skewed, with a median of 0 mm, indicating many dry days

```
    date           precipitation       temp_max        temp_min
Length:1461        Min.   : 0.000   Min.   :-1.60   Min.   :-7.100
Class :character   1st Qu.: 0.000   1st Qu.:10.60   1st Qu.: 4.400
Mode  :character   Median : 0.000   Median :15.60   Median : 8.300
                   Mean   : 3.029   Mean   :16.44   Mean   : 8.235
                   3rd Qu.: 2.800   3rd Qu.:22.20   3rd Qu.:12.200
                   Max.   :55.900   Max.   :35.60   Max.   :18.300
     wind          weather
Min.   :0.400   Length:1461
1st Qu.:2.200   Class :character
Median :3.000   Mode  :character
Mean   :3.241
3rd Qu.:4.000
Max.   :9.500
```

- Maximum temperature ranges from −1.6°C to 35.6°C, with an average of 16.4°C
- Minimum temperature ranges from −7.1°C to 18.3°C, with an average of 8.2°C
- Wind speed shows moderate variability, with a mean of 3.24
- The weather variable is categorical and describes daily weather conditions

Data quality with univariate analysis

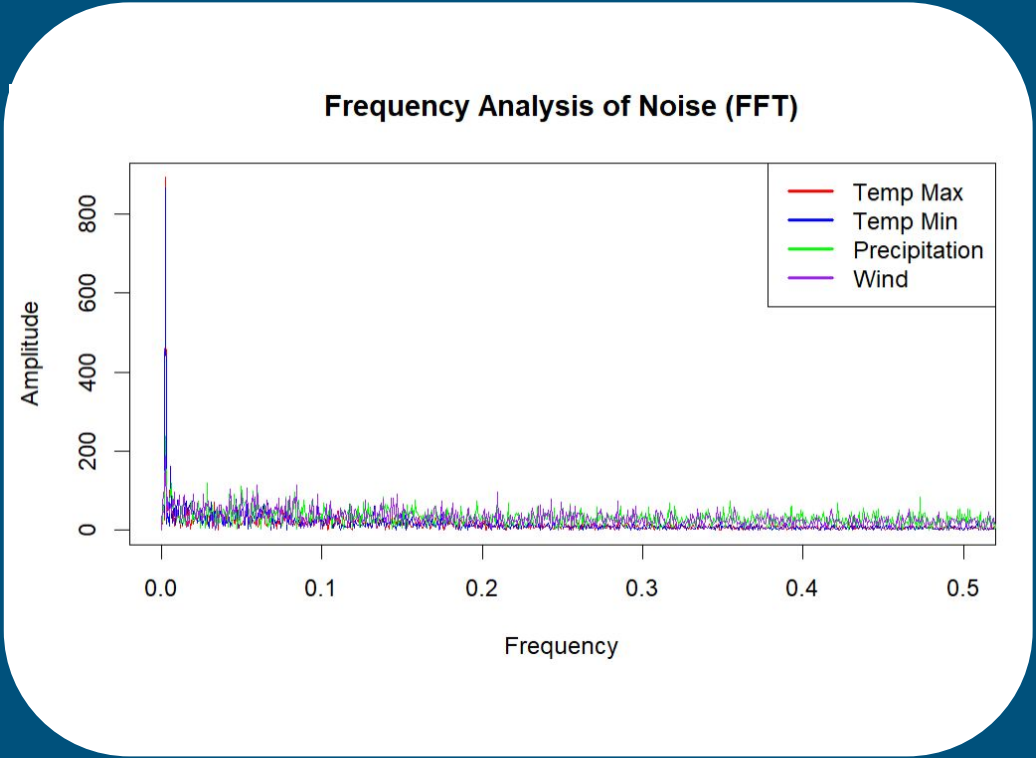
Noise:

This frequency spectrum represents the distribution of noise across different frequencies for each weather variable.

Low-frequency components correspond to slow variations, while high-frequency components represent rapid and irregular fluctuations.

The dominance of high frequencies indicates that the residual variability is mainly random noise rather than structured patterns.

By applying a Fourier Transform to the residuals, we identify whether the unexplained variability is dominated by slow trends or rapid random fluctuations.



Data quality with univariate analysis

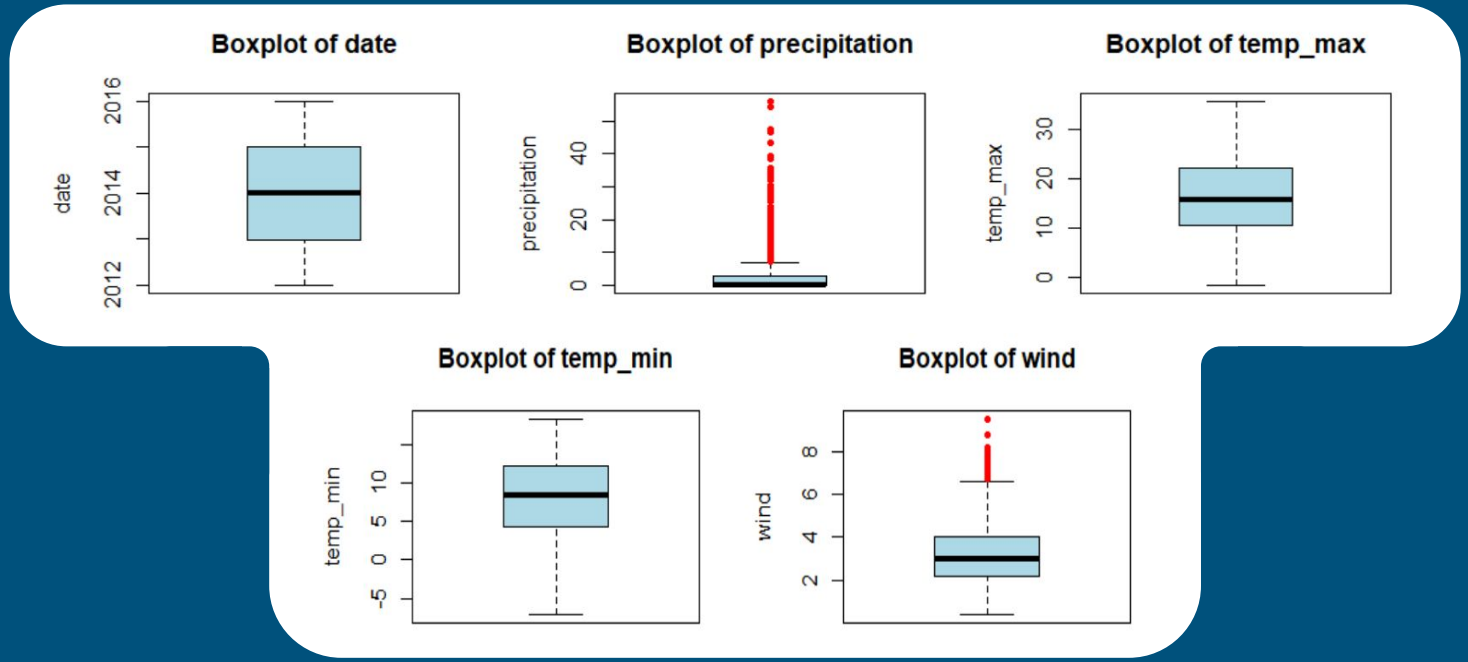
Outliers:

Outliers represent **extreme weather events** rather than errors.

They can provide **valuable information** about rare but significant temperature variations.

Data quality with univariate analysis

Outliers:



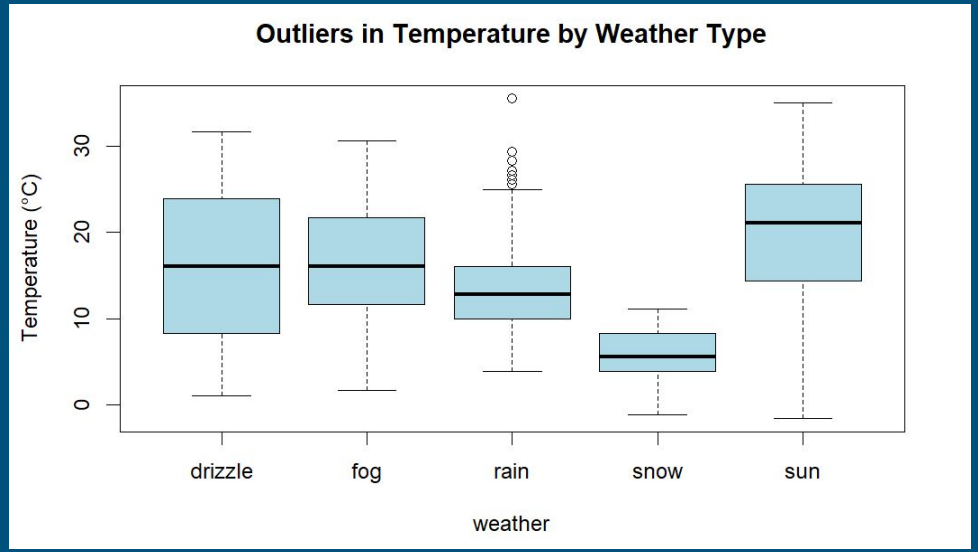
Data quality with univariate analysis

Outlier Detection:

- Rain shows the most temperature outliers (multiple points above 25°C), indicating unusual warm rainy days
- Sun has minimal outliers, suggesting consistent warm weather patterns
- Clear separation between winter conditions (snow) and summer conditions (sun)

Implications:

- Strong relationship between weather type and temperature
- Outliers in rainy weather may represent transition seasons (spring/autumn)
- Temperature can be a useful predictor for weather classification models



Data quality with univariate analysis

- <u>Missing values:</u>
- All variables were checked for missing values
- No missing values were detected in the dataset
- The dataset is complete and ready for analysis
- No imputation or data removal was required

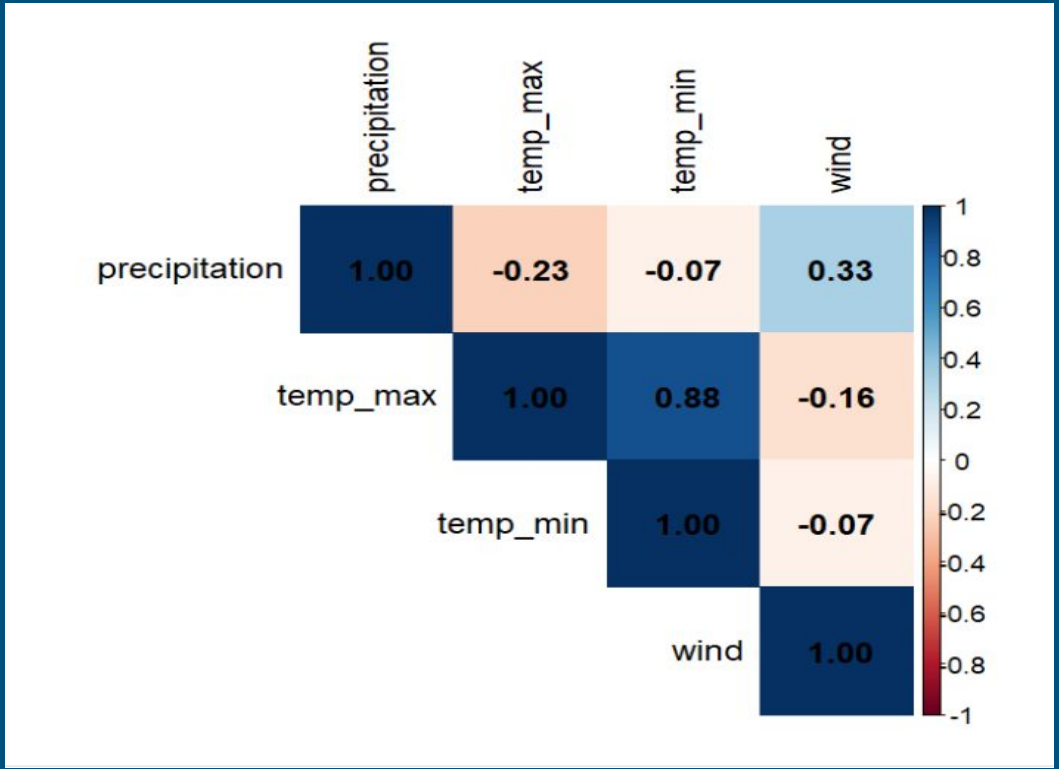
```{r}
weather$date <- as.Date(weather$date)
colSums(is.na(weather))
```

```
         date precipitation      temp_max      temp_min          wind       weather
            0             0             0             0             0             0
```

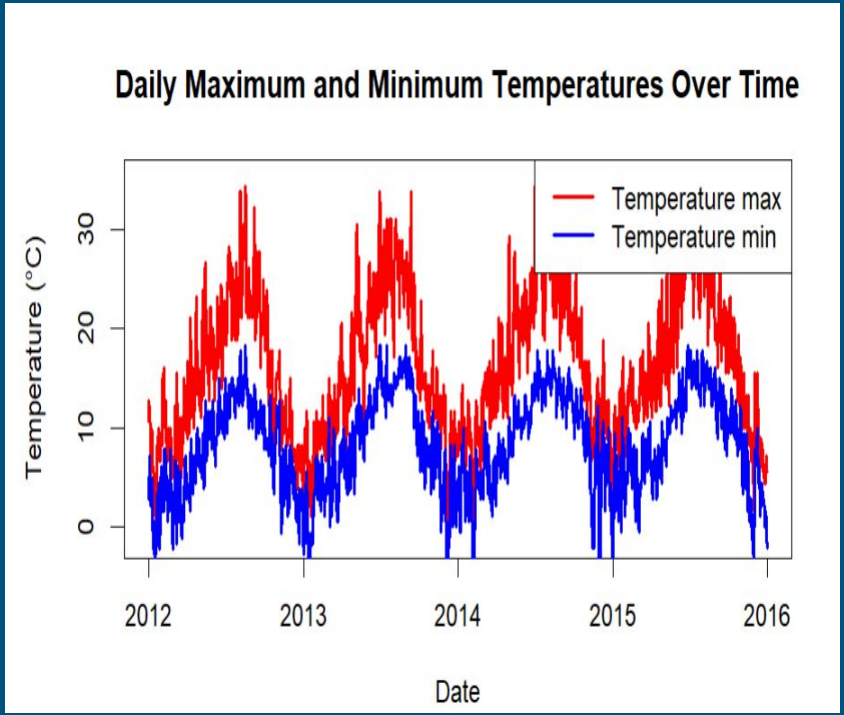
Dependences between the variables

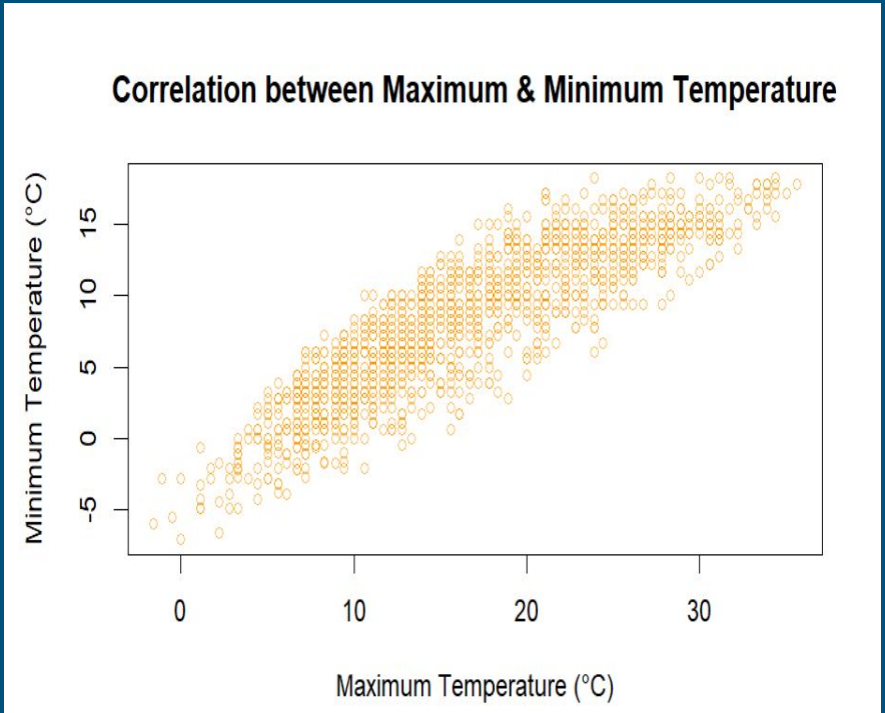
From the correlation matrix heatmap, it can be observed that the maximum and minimum daily temperatures are highly correlated, with a correlation coefficient of 0.88.

In contrast, the minimum daily temperature shows very weak correlations with wind speed and precipitation, with correlation coefficients of −0.07 and values close to zero, indicating that these variables are nearly independent.

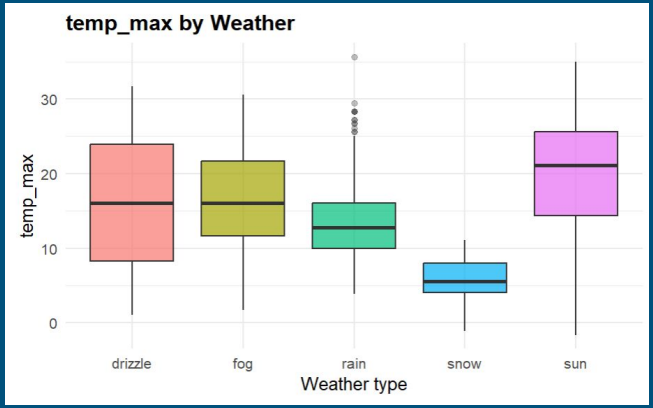


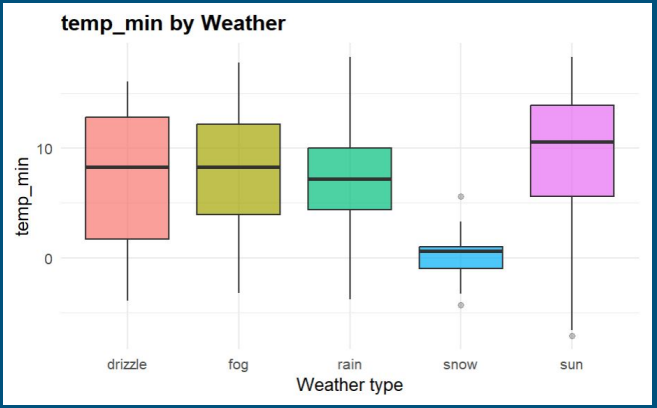
Dependences between the variables

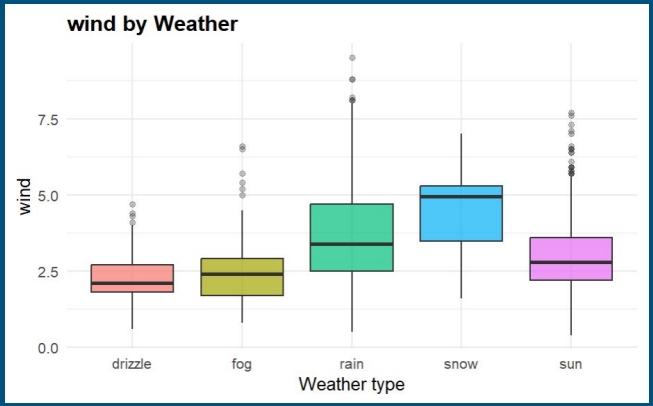


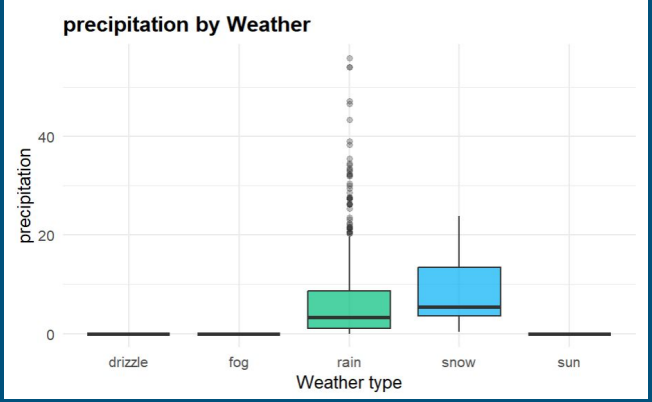


Multivariate statistical analysis of data









Selection of variables

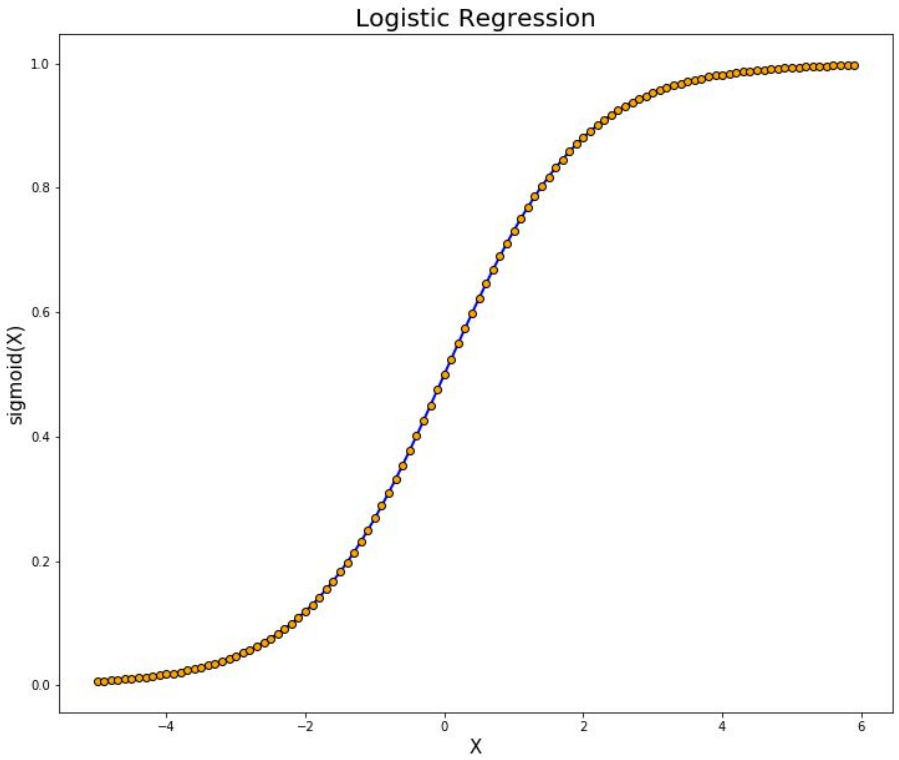
From the correlation plot between maximum and minimum temperatures, as well as the boxplots across weather categories, it is evident that these two variables carry very similar information. Therefore, we can select only one of them for further analysis.

Since wind speed and precipitation both show significant influence on weather classification, we retain these two features as well. In addition, temperature exhibits clear seasonal variation over time, so the time variable is naturally preserved.

Consequently, the final selected variables for our study are maximum temperature, wind speed, precipitation, and date.

Conclusion

Implementation of logistic regression



Reasons of choosing Softmax Method

Softmax is effective for weather classification because:

1. The problem is multiclass.
   Weather categories (sun, rain, fog, drizzle, etc.) are multiple classes, and softmax is designed to model the probability of each class.
2. It outputs probabilities.
   Softmax converts linear scores into probabilities between 0 and 1, and all probabilities sum to 1.
3. Classes compete with each other.
   If the probability of one weather type increases, the probabilities of the others decrease, which matches real-world situations where only one weather category occurs at a time.
4. It is a natural extension of logistic regression.
   Logistic regression works for binary classification, and softmax generalizes it to multiple classes.

Also, we don’t choose the Naive Bayes method because the conditional probability of features are not independent, so it would not be a good choice in this study.

So, softmax provides a mathematically consistent and interpretable way to predict weather categories from features like temperature, precipitation, wind, and month.

Data Preparation

As mentioned in the first part, the final selected variables for our study are maximum temperature, wind speed, precipitation, and date.

Specially, for the data of date, we converted into the data of month to better use the information included.

Besides, we have checked that we don’t have any missing values. As for the outliers, since the information in this case of outliers can indicate some extreme situations in weather, we decided to use all of the data.

Theoretical Explanation

Softmax Regression

Classical logistic regression is used for binary classification:

$$Y \in \{0, 1\}$$

with the log-odds model

$$\log \frac{P(Y = 1 \mid X)}{P(Y = 0 \mid X)} = w^T X$$

Thus,

$$\frac{P(Y = 1 \mid X)}{P(Y = 0 \mid X)} = e^{w^T X}$$

Using

$$P(Y = 0 \mid X) + P(Y = 1 \mid X) = 1$$

we obtain

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-w^T X}}$$

For multi-class classification,

$$Y \in \{1, 2, \ldots, K\}$$

choose class K as the reference class and assume

$$\log \frac{P(Y = k \mid X)}{P(Y = K \mid X)} = w_k^T X, \qquad k = 1, 2, \ldots, K-1$$

Then

$$\frac{P(Y = k \mid X)}{P(Y = K \mid X)} = e^{w_k^T X}$$

so

$$P(Y = k \mid X) = P(Y = K \mid X) e^{w_k^T X}$$

Using

$$\sum_{k=1}^{K} P(Y = k \mid X) = 1$$

Using

$$\sum_{k=1}^{K} P(Y = k \mid X) = 1$$

we get

$$P(Y = K \mid X) = \frac{1}{\sum_{j=1}^{K} e^{w_j^T X}}$$

hence

$$P(Y = k \mid X) = \frac{e^{w_k^T X}}{\sum_{j=1}^{K} e^{w_j^T X}}, \qquad k = 1, 2, \ldots, K$$

This is the Softmax model.

Likelihood

Given training data

$$(x_i, y_i), \qquad i = 1, 2, \ldots, n$$

define

$$y_{ik} = \begin{cases} 1, & y_i = k \\ 0, & y_i \neq k \end{cases}$$

Then the likelihood is

$$L(W) = \prod_{i=1}^{n} \prod_{k=1}^{K} P(Y = k \mid x_i)^{y_{ik}}$$

Taking the logarithm,

$$\ell(W) = \sum_{i=1}^{n} \sum_{k=1}^{K} y_{ik} \log P(Y = k \mid x_i)$$

Substituting the Softmax probabilities,

$$\ell(W) = \sum_{i=1}^{n} \sum_{k=1}^{K} y_{ik} \log \left(\frac{e^{w_k^T x_i}}{\sum_{j=1}^{K} e^{w_j^T x_i}} \right)$$

$$\ell(W) = \sum_{i=1}^{n} \sum_{k=1}^{K} y_{ik} \left(w_k^T x_i - \log \sum_{j=1}^{K} e^{w_j^T x_i} \right)$$

Equivalently, we minimize the negative log-likelihood

$$J(W) = -\ell(W)$$

$$J(W) = -\sum_{i=1}^{n} \sum_{k=1}^{K} y_{ik} \left(w_k^T x_i - \log \sum_{j=1}^{K} e^{w_j^T x_i} \right)$$

This is the multi-class cross-entropy loss.

Let

$$p_{ik} = P(Y = k \mid x_i) = \frac{e^{w_k^T x_i}}{\sum_{j=1}^{K} e^{w_j^T x_i}}$$

Then the gradient of the log-likelihood with respect to the parameter vector w_k is

$$\frac{\partial \ell(W)}{\partial w_k} = \sum_{i=1}^{n} x_i (y_{ik} - p_{ik})$$

If expanded to the feature level, the gradient with respect to each coefficient w_{kj} is

$$\frac{\partial \ell(W)}{\partial w_{kj}} = \sum_{i=1}^{n} x_{ij} (y_{ik} - p_{ik})$$

Equivalently, for the negative log-likelihood,

$$\frac{\partial J(W)}{\partial w_k} = \sum_{i=1}^{n} x_i (p_{ik} - y_{ik})$$

and

$$\frac{\partial J(W)}{\partial w_{kj}} = \sum_{i=1}^{n} x_{ij} (p_{ik} - y_{ik})$$

Iterative Optimization

Since there is no closed-form solution, the parameters are obtained using iterative optimization methods.

Using gradient ascent on the log-likelihood, the update rule at the feature level is

$$w_{kj}^{(t+1)} = w_{kj}^{(t)} + \eta \sum_{i=1}^{n} x_{ij}(y_{ik} - p_{ik})$$

where

$$p_{ik} = P(Y = k \mid x_i) = \frac{e^{w_k^T x_i}}{\sum_{j=1}^{K} e^{w_j^T x_i}}$$

Equivalently, minimizing the negative log-likelihood using gradient descent gives

$$w_{kj}^{(t+1)} = w_{kj}^{(t)} - \eta \sum_{i=1}^{n} x_{ij}(p_{ik} - y_{ik})$$

The parameters are updated iteratively until convergence.

Prediction

For a new sample X,

$$P(Y = k \mid X) = \frac{e^{w_k^T X}}{\sum_{j=1}^{K} e^{w_j^T X}}$$

and the predicted class is

$$\hat{y} = \arg\max_k P(Y = k \mid X)$$

Measure of Performance

To evaluate the performance of the Softmax classification model, we use a **confusion matrix** and **accuracy**.

The **confusion matrix** compares the predicted classes with the true classes and shows how many observations are correctly or incorrectly classified for each category.

The **accuracy** measures the proportion of correctly predicted observations among all observations. A higher accuracy indicates better classification performance of the Softmax model.